

SIMATS ENGINEERING

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Course Name: Cloud Computing and Big Data Analytics	Type: Operations Optimization
CO Statement:	Apply big data storage, integration, Hadoop, HDFS, and MapReduce concepts for distributed data processing

Part A: Production Hadoop Operations – Lesson Notes

Monitoring & Observability

A production Hadoop cluster generates thousands of metrics. Monitoring means: (1) collecting these metrics, (2) storing them, (3) displaying dashboards, (4) setting thresholds to trigger alerts.

Component	Key Metrics	Normal Range	Alert Threshold
NameNode	Heap usage, Live/Dead DataNodes, Block count, Edit log size	< 70% heap	> 85% heap
DataNode	Disk usage, Block count, Network I/O, JVM heap	< 80% disk	> 92% disk
MapReduce	Job completion rate, Task failure %, Shuffle sort time	< 5% failure	> 20% failure
HDFS	Under-replicated blocks, Missing blocks, Block replica imbalance	0 under-replicated	> 10 blocks

Lesson Note: Why Monitoring Matters

Without monitoring, you only discover problems when users complain. NameNode heap at 95% → crash → cluster down → data unavailable. Monitoring alerts at 85% gives you time to act (add DataNodes, scale up, optimize)

Backup & Disaster Recovery

NameNode stores the file system namespace (file names, hierarchy, replica locations) in memory and on disk (fsimage + edit logs). If NameNode crashes and no backup exists, you lose the entire namespace — irrecoverable.

Key Concepts:

1. RTO (Recovery Time Objective): How long can the cluster be down? 1 hour? 1 minute?
2. RPO (Recovery Point Objective): How much data loss is acceptable? 1 hour? 1 day?

Backup Strategy	RTO	RPO	Cost	Best For
No backup (risky)	∞ (unrecoverable)	∞	\$0	Dev/test only
Daily full backup to external disk	1–2 hours	24 hours	\$100/month	Dev, non-critical
Hourly incremental + offsite replication	15 minutes	1 hour	\$500/month	Production
HA (active/standby NameNode + ZK quorum)	< 1 minute	0 (near-instant)	\$1000+/month	Critical production

High Availability (HA) Design

HA eliminates the single NameNode as a single point of failure. Setup: Active NameNode + Standby NameNode + ZooKeeper quorum (3 nodes). When Active fails, Standby takes over automatically in < 1 minute. Requires shared storage (NFS/QJM) for edit logs.

Cost Optimization

Monthly cluster cost = Hardware + Power + Cooling + Ops

Cost Component	Typical \$	Control Method
Hardware (50 nodes × \$500)	\$25,000 one-time	Buy less, or decommission slow nodes
Annual power & cooling (50 nodes × \$1000)	\$50,000/year	Efficient hardware, underutilization
Operations & support	\$10,000/month	Automation, monitoring, runbooks
Compression (Gzip, Snappy)	\$-5,000/month savings	Reduced storage, higher CPU
Tiering (move old logs to cold storage)	\$-3,000/month savings	Archive logs > 30 days to S3

Scaling Strategies

Horizontal scaling: Add more DataNodes (linear cost, complexity)

Vertical scaling: Upgrade existing nodes (cap at 10–20TB/node practical limit)

NameNode Capacity Constraint

NameNode heap number of files. Rule of thumb: 1GB heap ≈ 1M files. If 100M files needed, you need 100GB NameNode heap — impractical. Solutions: (1) Tiering (archive old data), (2) Federation (multiple NameNodes), (3) Consolidate small files.

Part B: Main Scenario – Optimize a 50-Node Production Cluster

Current State Assessment

A company runs a 50-node HDFS cluster processing 10TB/day. Current metrics:

Metric	Current Value	Status
NameNode heap usage	85%	RISKY – near crash threshold
DataNode disk utilization	92%	RISKY – no space for replication
Monthly cost	\$50,000	High, no optimization
Last backup	2 weeks ago	UNACCEPTABLE RPO
DataNode failure recovery time	8 hours	UNACCEPTABLE RTO
Monitoring setup	Manual spreadsheets	No real-time alerts
HA setup	None (single NameNode)	Single point of failure

Problem Summary

The cluster is on the edge of failure. NameNode heap crash is imminent. No automated failover. Slow recovery from node failures. Costs unchecked. Backup is stale.

Step 1: Design Monitoring Dashboard

Create a monitoring strategy: what to track, alert thresholds, and dashboard design.

1a. NameNode Metrics

Metrics to track:

- Heap usage (%)
- Live/Dead DataNodes (count)
- Block count (total)
- Edit log size (MB)

Metric	Green	Yellow	Red
NameNode heap %	< 60%	60–80%	> 80%
Dead DataNodes	0	1–2	> 2
Under-replicated blocks	0	< 100	> 100

Fill in your thresholds and actions:

Track NameNode heap, dead DataNodes and under-replicated blocks continuously. Use Green/Yellow/Red thresholds from the table: heap <60% / 60–80% / >80%; dead DataNodes 0 / 1–2 / >2; under-replicated blocks 0 / <100 / >100. Red alerts should trigger immediate investigation, node recovery, rebalancing and re-replication. Display current value, trend, alert state and action on the dashboard.

Step 2: Plan Backup & Disaster Recovery

Current state: Last backup 2 weeks ago → RPO of 14 days is unacceptable.

2a. Backup Strategy

Design a backup plan:

Decision	Option A	Option B	Option C
Backup frequency	Daily	Hourly	Real-time (HA)
Storage location	On-cluster NFS	Off-site S3	Shared QJM
RTO	2–4 hours	1 hour	< 1 minute
RPO	24 hours	1 hour	Near-zero
Cost/month	\$200	\$1000	\$2000+ HA infra

Your recommendation: Choose the option that balances cost and business needs.

Choose Option B — hourly incremental backup with off-site S3. It gives RPO = 1 hour and RTO = 1 hour for \$1,000/month, making it a practical production balance. Daily backup is cheaper but allows 24 hours of data loss; real-time HA is faster but costs more.

Step 3: Cost Optimization Analysis

Current: 50 nodes × \$1000/year ops = \$50K/year. Target: Reduce by 20% to \$40K/year.

Optimization	Annual Savings	Impact	Your Choice?	
Compression (Gzip logs)	\$5,000	Reduced storage, higher CPU	Yes / No	YES
Tiering (archive > 30 days to cold)	\$3,000	Slower access to old logs	Yes / No	YES
Decommission 5 slow nodes	\$5,000	Less capacity, need to rebalance	Yes / No	YES
Auto-scaling (scale down off-peak)	\$2,000	Reduced peak hours ops	Yes / No	YES

Calculate total savings from your selections and new monthly cost:

Selected savings = \$5,000 + \$3,000 + \$5,000 + \$2,000 = \$15,000/year. Current annual cost = \$50,000, so new annual cost = \$35,000/year. This is a 30% reduction and exceeds the \$40,000/year target. Decommission slow nodes only after rebalancing.

Step 4: Address NameNode Heap Pressure

Current heap: 85% at 50M files. At current growth (1M files/month), heap will exceed 90% in ~2 months → crash risk.

Option	Action	NameNode Heap Impact	Timeline
Tiering	Archive logs > 30 days (removes old files)	Reduces heap by ~10%	1 month
Consolidate	Merge small files into larger blocks	Reduces heap by ~15%	Ongoing
Scale	Add second NameNode (Federation)	Splits namespace across 2 NameNodes	2 months setup

Which solution do you recommend? Why?

Recommend Consolidation first because it reduces NameNode heap pressure by about 15% and directly addresses the large file count. Also use Tiering to archive logs older than 30 days and reduce heap by about 10%. Use Federation as the longer-term scale-out solution if file growth continues.

Step 5: Implement High Availability (HA)

Current: Single NameNode (SPOF). Proposed: HA with Standby.

Component	Quantity	Purpose
Active NameNode	1	Primary metadata server
Standby NameNode	1	Automatic failover replica
ZooKeeper quorum	3 or 5	Coordinates HA election
Shared edit log storage	NFS or QJM	Both NNs access same metadata

Failover process: Active NameNode fails → ZK detects (30 sec) → Standby promotes (30 sec) → Clients reconnect.
Total RTO < 1 minute.

Estimate HA setup cost and timeline:

Estimated HA setup cost: about \$2,000/month for HA infrastructure, based on the workbook's HA cost range. Allow about 2–4 weeks for configuration, shared edit-log setup, ZooKeeper quorum, failover testing and production rollout. The target RTO is less than 1 minute.

Part C: Mini-Problems

Mini-Problem 1: DataNode Failure Scenario

A DataNode holding 12TB of data fails (hardware crash). HDFS detects the failure and begins re-replicating blocks from surviving replicas. Current network capacity: 1 Gbps per DataNode.

Calculate re-replication time and estimate when cluster is back to normal replication factor.

12 TB = 96 Tb. At 1 Gbps, ideal transfer time = 96,000 seconds $\approx$ 26.7 hours. Allow extra time for disk, network and protocol overhead; a practical estimate is about 28–32 hours. Normal replication is restored after all affected blocks are re-replicated.

Mini-Problem 2: NameNode Heap Projection

Current state: 50M files, 60% heap usage. Growth rate: 1.5M files/month. If file count grows linearly, when will NameNode heap exceed 90%? When should you add a second NameNode (Federation)?

50M files use 60% heap. Assuming linear scaling, 90% heap corresponds to 75M files. Growth needed = 25M files; at 1.5M/month, $25 \div 1.5 \approx 16.7$ months. Heap will exceed 90% in about 17 months. Add Federation around 12–15 months from now as a safety margin.

Mini-Problem 3: Compression Trade-off

Implement Gzip compression on 20TB of historical logs. Storage reduction: 3:1 (20TB $\rightarrow$ 6.67TB). But: CPU overhead for decompression during MapReduce jobs increases job time by 15%.

Is compression worth it economically? Calculate storage cost savings vs. increased compute cost.

Compression reduces 20 TB to 6.67 TB, saving 13.33 TB. The supplied data does not give storage or compute prices, so an exact dollar saving cannot be calculated. Compression is economically worthwhile when the monthly storage saving from 13.33 TB exceeds the extra compute cost caused by the 15% longer jobs.

Mini-Problem 4: Tiering Strategy

Move logs older than 30 days to cold storage (AWS S3 Glacier, \$1/TB/month). Current hot storage cost: \$50/TB/month. Hot tier capacity: 20TB. Cold tier capacity: unlimited.

Calculate break-even point: after how many days of cold storage does it become cheaper than hot?

Hot storage costs \$50/TB/month while Glacier costs \$1/TB/month. For 1 TB, the saving is \$49 per month. Cold storage is therefore cheaper immediately after transfer; the break-even point is effectively day 1, ignoring one-time transfer and request charges.

Part D: Problem Bank – 4 Additional Scenarios

Scenario A: Multi-Datacenter Failover

Company wants to replicate the 50-node cluster to a second datacenter 100km away for disaster recovery. Design cross-datacenter replication strategy, network bandwidth requirements, and cost implications. Should data be synchronously or asynchronously replicated? What is RTO/RPO?

Use asynchronous cross-datacenter replication for disaster recovery. For 10 TB/day, average generation is about 116 MB/s, so provision at least 1 Gbps sustained replication bandwidth; 2 Gbps or more is preferable for overhead and recovery backlog. Target RPO about 1 hour and RTO about 15–60 minutes. Synchronous replication would add write latency and requires a low-latency link.

Scenario B: Federation for Scale-Out

The cluster needs to support 200M files (not possible with single NameNode due to heap limits). Design a Federation setup: how many NameNodes? How to distribute files across namespaces? What is the performance impact? Can you still run cluster-wide MapReduce jobs?

Use HDFS Federation with 3 NameNodes, giving roughly 67M files per namespace for 200M files. Distribute directories or tenants across namespaces using a routing policy. Federation reduces per-NameNode heap pressure and increases namespace capacity. Cluster-wide MapReduce can still run because compute resources remain shared.

Scenario C: Real-Time Alerting & Auto-Remediation

Design a monitoring + auto-remediation system: when DataNode disk usage exceeds 85%, automatically trigger: (1) alert to ops, (2) pause new block writes to that node, (3) trigger re-replication of critical blocks. What tools would you use? How to test safely?

Use Prometheus/Grafana with Alertmanager, plus Ansible or an automation service for remediation. When disk usage exceeds 85%, alert operations, stop or limit new writes to that node, and trigger re-replication of critical blocks. Test in staging with synthetic data: raise disk usage, verify the alert, confirm write throttling and re-replication, then test rollback before production.

Scenario D: Cost Benchmarking & Optimization

The company wants to understand their true cost per TB stored. Include: hardware amortization, power, cooling, network, ops salary (allocate 10% of an engineer's time). Compare: (1) on-premise cluster, (2) AWS S3, (3) hybrid (hot on-premise + cold cloud). Which is most cost-effective for 100TB with 2-year horizon?

Build a two-year cost-per-TB model including hardware amortization, power, cooling, network and 10% of an engineer's salary. For 100 TB, compare on-prem HDFS, AWS S3 and hybrid hot on-prem + cold cloud. On-prem is usually cheapest at high utilization; S3 offers flexibility; hybrid often gives the best balance for hot recent data and cold historical data. Exact dollar comparison requires the missing local price inputs.

Evaluation Rubric

Criteria	Wt.	L4 (40–50)	L3 (30–39)	L2 (20–29)	L1 (0–19)
Monitoring & Observability	20%	Comprehensive metrics, realistic alert thresholds, actionable	Good metrics; some thresholds arbitrary	Basic metrics; incomplete alert strategy	Incomplete monitoring design
Cost Optimization	25%	Accurate cost breakdown, realistic savings targets, trade-offs justified	Good analysis; minor gaps in cost modeling	Basic cost analysis; limited optimization ideas	Incorrect or missing cost analysis
Backup & DR Planning	20%	Credible RTO/RPO targets, feasible failover design, HA understanding	Good DR plan; some RTO/RPO unrealistic	Basic backup strategy; weak failover plan	Incomplete or missing strategy
Scaling & Capacity Planning	20%	Data-driven forecasts, proactive scaling, Federation/HA impact analysis	Good capacity plan; some growth projections weak	Basic capacity analysis; limited forward planning	Missing or unrealistic capacity planning
Documentation & Clarity	15%	Clear dashboards, formulas, step-by-step reasoning, justified decisions	Mostly clear; minor formatting/detail gaps	Understandable but lacks design diagrams/detail	Unclear or poorly organized

Total Marks: 100 (20 + 25 + 20 + 20 + 15)